

AI Friday National Finals Season 2

Consolidated Use Case Portfolio: 20 Challenges

Source-faithful Markdown transcription of the uploaded portfolio. The content below preserves the challenge wording and structure as closely as possible. Formatting artifacts from the source PDF have been normalized into Markdown headings, lists, and tables. No ratings, tool recommendations, or additional solution suggestions have been added.

01. Card Dispute Evidence Reconstruction & Resolution Agent

Business Context & Why It Matters

Disputes contain conflicting customer, merchant and system narratives, and important facts can arrive late after the case has already progressed.

The operating environment spans customer statements, merchant records, transaction events, receipts, delivery records, authentication events and correspondence. Evidence can arrive asynchronously, be corrected later or conflict across systems, so the solution must preserve state, provenance and prior actions as the case evolves.

Complexity layer: The solution must reconcile changing evidence across multiple systems, maintain long-running state, support real approvals and continue safely through partial failures. Material changes must trigger targeted re-evaluation rather than silently preserving an outdated conclusion.

The Challenge & Scope

Cards & Payments · Payments Operations · Evidence Intelligence

What Teams Must Reimagine

Reconstruct a trustworthy disputed event, expose gaps and contradictions, and recommend the next useful evidence or permitted case action.

The outcome should be an evidence-backed operational workspace that can progress the journey, surface uncertainty and coordinate authorized actions while preserving accountability. Final liability and consequential case decisions remain human-owned.

Core Solution Capabilities

Your solution should be able to:

- Assemble and reconcile customer statements, merchant records, transaction events, receipts, delivery records, authentication events and correspondence into a coherent, time-aware operating view.

- Detect missing, stale, duplicated or contradictory information and make uncertainty explicit.
- Dynamically select the next useful evidence, analysis, coordination or permitted action based on the current state and constraints.
- Maintain source provenance, human intervention, approvals and a complete audit trail through the full journey.

End-to-End Capability Expectations & Controls

- Build a durable domain model that links customer statements, merchant records, transaction events, receipts, delivery records, authentication events and correspondence and can represent late, corrected or conflicting versions without losing earlier evidence.
- Determine the next-best evidence request or permitted action dynamically from current state, deadlines, dependencies, authority and expected value rather than relying on a fixed happy path.
- Track long-running workflow state, deadlines and cross-system dependencies while preventing duplicate requests, duplicate transactions or repeated external actions.
- Represent competing interpretations, plans or hypotheses and show which evidence supports, weakens or changes each option without hiding uncertainty.
- Maintain versioned state with source-level provenance and explicit separation of recorded fact, AI inference, user input, automated action and human decision.
- Coordinate approved tools or specialist AI components with permission checks, retries, timeouts, idempotency and safe recovery when only part of a workflow succeeds.
- Support role-based intervention, approval gates, audit replay and configurable action boundaries so reviewers can reconstruct what the system knew, why it acted and what changed afterward.

Production-Ready Solution & Data

Environment

Build a production-ready, end-to-end working solution engineered for enterprise deployment and operational use. The solution should demonstrate production-grade architecture, security, resilience, observability, auditability, configurable controls, failure recovery and maintainable integration patterns. Where live enterprise, partner or external-system access is unavailable, teams may use representative mock interfaces or simulated services, but the solution itself must behave as a deployable product.

It should ingest representative customer statements, merchant records, transaction events, receipts, delivery records, authentication events and correspondence, execute the complete journey, react safely to late-arriving or changed information, recover from partial failure and retain a complete inspectable audit trail. Final liability and consequential case decisions remain human-owned.

Data Availability & Synthetic Data

Use synthetic starter data where relevant. Teams may create additional synthetic customer statements, merchant records, transaction events, receipts, delivery records, authentication events and correspondence plus representative edge cases, corrections, failure responses and approval states needed to demonstrate the complete solution. No live confidential enterprise data is required.

End-to-End Journey

1. Dispute raised
2. Evidence gathered
3. Event reconstructed
4. Narratives compared
5. Gaps identified
6. Evidence requested
7. Interpretation prepared

8. Specialist reviewed
9. Resolution progressed

National Finale Inject

A merchant provides material evidence late, contradicting the customer's original description. The solution must incorporate it, reassess dependent conclusions and make the change visible.

02. AI-Powered Supply Chain Disruption Planning & Recovery Engine

Business Context & Why It Matters

Supply chains operate through interconnected planning and operational processes spanning suppliers, plants, warehouses, transportation and customers. Disruptions can invalidate an approved plan quickly, while local fixes can create downstream shortages, service failures or excess cost.

The challenge is to build an intelligent decision platform that understands the current operating state, identifies material exceptions, evaluates their downstream impact and recommends feasible recovery actions before execution changes are committed.

Complexity layer: The solution must reconcile changing operational evidence, constraints and dependencies across a multi-node network, maintain state over a long-running recovery workflow, preserve provenance and approvals, and recover safely from partial execution or

changed conditions.

The Challenge & Scope

Manufacturing / Supply Chain · Operational Decision Intelligence · Simulation-Backed Recovery

Synthetic Data Supported · Laptop-Executable · Production-Ready Working Solution · No Participant-Built Forecasting Required

Forecasting and machine-learning model building are removed as the primary participant deliverable. Baseline demand / forecast signals are provided as challenge inputs. The core challenge is now the closed operational loop:

Plan → Monitor → Detect → Investigate → Generate Alternatives → Simulate → Recommend → Human Approve → Re-plan

What Teams Must Reimagine

Build a production-ready AI solution that starts from a feasible baseline plan and continuously monitors execution, detects disruption, investigates affected dependencies, generates alternative responses, evaluates those alternatives through simulation and recommends the best recovery path.

The solution should balance service, cost, inventory, capacity and business priorities while keeping planner approval as the final control for material plan changes.

Core Solution Capabilities

- Ingest baseline demand / forecast signals, sales and orders, supplier commitments, capacity calendars, inventory, transport lanes and operational status.

- Maintain a time-aware representation of plants, suppliers, warehouses, customers and transport dependencies.
- Detect structured and unstructured disruptions and identify affected products, orders, facilities, lanes and customer commitments.
- Explain the causal chain from the disruption to operational and business impact.
- Generate feasible recovery alternatives such as alternate suppliers, rerouting, production reallocation, inventory redeployment, prioritisation and expediting.
- Simulate the operational consequences of alternatives against constraints and selected business objectives.
- Recommend a recovery option with clear evidence, trade-offs, assumptions and confidence.
- Support planner approval, revised-plan generation, execution tracking and an auditable record of what changed.

End-to-End Capability Expectations & Controls

- **Operational state model:** Represent the current and planned state of suppliers, plants, warehouses, inventory, orders, transport and customer commitments, including late or corrected information.
- **Exception investigation:** Connect events across the network and distinguish root causes from secondary symptoms.
- **Constraint-aware recovery:** Ensure proposed actions respect capacity, supply, transport, inventory and service constraints.
- **Scenario generation:** Create multiple editable response options rather than a single hard-coded recovery.
- **Simulation-backed evaluation:** Compare service, cost, inventory, utilisation, timing and other selected objectives before commitment.
- **Dynamic prioritisation:** Re-evaluate recommendations when material facts, constraints or priorities change.
- **Safe orchestration:** Enforce permissions, approvals, idempotency, retries, timeouts and partial-failure recovery for

external or simulated actions.

- **Traceability:** Keep source provenance and distinguish recorded facts, AI inferences, user decisions, simulated outcomes and committed actions.

Production-Ready Solution & Data Environment

Build a deployable end-to-end working solution using synthetic or challenge-provided data. Teams are not expected to develop a sophisticated demand-forecasting model. Baseline demand / forecast inputs will be available as part of the challenge data.

The solution should demonstrate network-state understanding, disruption detection, investigation, alternative generation, simulation, recommendation, human approval and revised-plan execution. Representative mock interfaces or simulated services are permitted where live enterprise systems are unavailable, but security, resilience, observability, auditability and maintainability remain part of the expectation.

Data Availability & Synthetic Data

Teams may use synthetic, open or self-generated data. The demonstration should contain a realistic multi-product network with suppliers, plants, warehouses, customers and transport links, plus baseline plans and deliberately introduced disruptions. Include structured operational data and representative communications such as emails or exception notices. No live enterprise data or live supply-chain platform access is required.

End-to-End Journey

1. Baseline plan loaded

2. Execution monitored
3. Disruption detected
4. Impact investigated
5. Recovery options generated
6. Scenarios simulated
7. Recommendation prepared
8. Planner approved
9. Plan revised and tracked

National Finale Inject

A critical supplier announces a seven-day delay affecting two components. At the same time, a priority customer increases demand by 18% and an alternate transport lane becomes constrained. The solution must determine the affected network, generate feasible recovery options, simulate the trade-offs, recommend the best path and prevent conflicting or duplicate reallocations.

03. Procurement Savings Assurance Solution

Business Context & Why It Matters

Procurement savings are often constrained before negotiation starts. Requirements may contain excess quantities or unnecessarily premium specifications, category strategies may rely on incomplete market context, supplier bids may use different units and pricing structures, and negotiated value can be weakened when translated into contract language.

The opportunity is to build a connected AI solution that challenges

demand, improves sourcing and price outcomes, and protects the negotiated value until contract signature.

The Challenge & Scope

Procurement · Source-to-Contract · Commercial Decision Intelligence

Build a production-ready AI solution that starts with an initial procurement requirement and ends with a signed-contract-ready commercial position. The solution should reason across requirements, consumption and spend data, market signals, known supplier bids, negotiation positions and contract documents.

Every material recommendation must be traceable to evidence and high-impact or ambiguous decisions must be routed for human review.

1. Challenge and Optimise the Demand

- Test requested volumes against historical consumption, forecasts and actual need.
- Identify duplicate, fragmented or overlapping requests that can be consolidated.
- Assess whether quantities can be reduced, phased or timed differently.
- Challenge unnecessary buffers, minimums or contingency volumes.
- Simplify materials, features, brands, technical requirements or service levels where appropriate.
- Assess substitutes, alternative designs or equivalent offerings.
- Identify specifications that unnecessarily restrict competition or raise cost.
- Surface supplier-led innovation, alternative commercial models and lifecycle-cost improvements.

2. Get the Price Right

- Use spend and market signals to determine a defensible category approach.
- Develop an independent should-cost or fair-price position using relevant cost drivers, volumes, benchmarks and transparent assumptions.
- Explain why the commercial position is economically defensible.
- Assume a known supplier pool and orchestrate an RFX around the optimised requirement.
- Normalise bids across currencies, units, bundles, tiers and commercial terms.
- Identify hidden costs and conditions and evaluate total value rather than headline price.
- Establish negotiation targets, levers, trade-offs, fallbacks and walk-away positions.
- Recommend a negotiation position or controlled next action; bounded automation may be demonstrated within clear guardrails.

Synthetic Supplier Bids

Teams should create or use deliberately non-normalised supplier bids with varied currencies, units, bundles, tiers, hidden costs and commercial conditions. The solution must perform the normalisation and total-value reasoning rather than receive a pre-cleaned comparison.

3. Lock the Value into a Tight Contract

- Translate negotiated commercial positions into appropriate contract clauses and schedules.
 - Draft tight, standards-aligned language without inventing unsupported obligations or terms.
 - Compare supplier redlines with the agreed commercial position.
 - Explain the savings or value implication of material redlines.
 - Identify language that weakens pricing, rebates, service levels, commitments or protections.
 - Route ambiguous or high-impact changes for human review.
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Core Solution Capability

Maintain one traceable thread from the original requirement and category strategy through supplier bids, negotiated position and contract clauses.

Evidence chain: Requirement → Optimised Need → Category Strategy → Should-Cost → RFx/Bids → Negotiated Position → Contract Language → Value Assurance

Production-Ready Solution & Data Environment

The core source-to-contract journey is retained. Teams are expected to demonstrate negotiation strategy and controlled recommendations; fully autonomous negotiation is not mandatory. The required journey ends at contract signature/value assurance, with post-signature compliance and procure-to-pay outside scope.

Teams may use synthetic, open or self-generated information. No real client or supplier-confidential data is required.

Representative data includes:

- Demand and consumption history
- Spend and category data
- Market and commodity signals
- Known supplier profiles
- RFx material and non-normalised bids
- Negotiation positions and fallback rules
- Contract templates and supplier redlines

Build a deployable end-to-end working solution across demand optimisation, strategy, sourcing, price analysis, negotiation recommendation and contract value assurance. The solution should demonstrate grounding, provenance, human review, security, resilience, observability, auditability, maintainability and cost-to-scale

considerations. Live supplier systems, procurement platforms or contract repositories are not required; representative mock interfaces are acceptable.

End-to-End Journey

1. Requirement ingested
2. Demand challenged
3. Category strategy set
4. Should-cost established
5. RFx and bids evaluated
6. Negotiation position prepared
7. Contract drafted
8. Redlines challenged
9. Value assured

National Finale Inject

Supplier B initially appears to offer the lowest total price. A late contract redline introduces a minimum-volume commitment, revised indexation mechanism and logistics exclusion that materially change the economics. The solution must trace the redline back to the negotiated position and determine whether the supplier remains the preferred commercial option.

04. AI-Powered Dynamic Covenant Monitoring & Early Warning

Business Context & Why It Matters

Commercial lending covenants are often monitored through periodic borrower reporting and manual review. A borrower can deteriorate between reporting cycles, while warning signals may already exist across account activity, utilization, payments, treasury flows, industry conditions and concentration exposure.

Earlier detection can help relationship, credit and risk teams intervene before a formal breach occurs.

The Challenge & Scope

Design and build a production-ready AI solution that monitors contractual covenant thresholds together with borrower financial and behavioural signals, forecasts breach risk over 30-, 60- and 90-day horizons, explains the drivers of deterioration, and recommends prioritized interventions.

Core challenge: The system must distinguish meaningful deterioration from temporary noise and provide evidence-backed early warning rather than merely flagging a covenant after it has already been breached.

Core Solution Capabilities

- Ingest borrower financial statements and calculate relevant financial ratios.
- Extract and represent covenant definitions, thresholds, testing frequency and exceptions.
- Monitor account activity, payment behaviour and credit/facility utilization.
- Evaluate treasury flows and changes in cash movement patterns.
- Incorporate concentration exposure and synthetic industry/news deterioration signals.
- Predict probability of covenant breach at 30-, 60- and 90-day

horizons.

- Identify the primary drivers contributing to forecast risk.
- Rank borrowers/facilities by urgency and expected impact.
- Recommend appropriate relationship-manager, credit or risk-team interventions.
- Generate an auditable warning trail showing data, trends, calculations and reasoning.

Early-Warning Intelligence

The solution should combine deterministic covenant calculations with predictive indicators. Examples include:

- Weakening debt-service or leverage position
- Rapid utilization increase
- Delayed or irregular payments
- Deteriorating cash inflows
- Industry stress
- Concentration risk
- Unusual treasury-flow changes

Production-Ready Solution & Data Environment

Teams should create synthetic commercial borrower portfolios with financials, covenants, account activity, payment history, facility utilization, industry indicators and treasury flows.

- No live core-banking or credit system integration is required.
- External news/industry signals may be supplied or synthetically generated.
- The full scoring and alerting workflow should be executable locally.

End-to-End Journey

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1. **Borrower and Covenant Intake:** Load financials, facilities, covenant definitions and thresholds.
 2. **Signal Monitoring:** Track payments, utilization, account activity, treasury and external risk indicators.
 3. **Risk Forecast:** Estimate breach probability at 30, 60 and 90 days.
 4. **Driver Explanation:** Explain which signals and covenant movements are creating risk.
 5. **Portfolio Prioritization:** Rank borrowers by urgency, confidence and potential exposure.
 6. **Intervention:** Recommend actions and maintain an auditable early-warning history.
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05. Autonomous Product Intelligence Factory

Business Context & Why It Matters

Retail product data arrives from suppliers in inconsistent formats, while channel, taxonomy and compliance requirements vary by market.

The operating environment spans supplier spreadsheets, documents, images, taxonomy, channel rules, regulatory attributes and correction feeds. Evidence can arrive asynchronously, be corrected later or conflict across systems, so the solution must preserve state, provenance and prior actions as the case evolves.

Complexity layer: The solution must reconcile changing evidence across multiple systems, maintain long-running state, support real approvals and continue safely through partial failures. Material changes must trigger targeted re-evaluation rather than silently preserving an outdated conclusion.

The Challenge & Scope

Convert fragmented supplier inputs into governed, source-linked and channel-ready product records.

The outcome should be an evidence-backed operational workspace that can progress the journey, surface uncertainty and coordinate authorized actions while preserving accountability. Regulated claims and material product assertions require configured human or policy approval.

Core Solution Capabilities

- Assemble and reconcile supplier spreadsheets, documents, images, taxonomy, channel rules, regulatory attributes and correction feeds into a coherent, time-aware operating view.
- Detect missing, stale, duplicated or contradictory information and make uncertainty explicit.
- Dynamically select the next useful evidence, analysis, coordination or permitted action based on the current state and constraints.
- Maintain source provenance, human intervention, approvals and a complete audit trail through the full journey.

End-to-End Capability Expectations & Controls

- Build a durable domain model that links supplier spreadsheets, documents, images, taxonomy, channel rules, regulatory attributes and correction feeds and can represent late, corrected or conflicting versions without losing earlier evidence.
- Determine the next-best evidence request or permitted action dynamically from current state, deadlines, dependencies, authority and expected value rather than relying on a fixed happy path.
- Track long-running workflow state, deadlines and cross-system

dependencies while preventing duplicate requests, duplicate transactions or repeated external actions.

- Represent competing interpretations, plans or hypotheses and show which evidence supports, weakens or changes each option without hiding uncertainty.
- Maintain versioned state with source-level provenance and explicit separation of recorded fact, AI inference, user input, automated action and human decision.
- Coordinate approved tools or specialist AI components with permission checks, retries, timeouts, idempotency and safe recovery when only part of a workflow succeeds.
- Support role-based intervention, approval gates, audit replay and configurable action boundaries so reviewers can reconstruct what the system knew, why it acted and what changed afterward.

Production-Ready Solution & Data Environment

Build a production-ready, end-to-end working solution engineered for enterprise deployment and operational use. It should ingest representative supplier spreadsheets, documents, images, taxonomy, channel rules, regulatory attributes and correction feeds, execute the complete journey, react safely to late-arriving or changed information, recover from partial failure and retain a complete inspectable audit trail. Regulated claims and material product assertions require configured human or policy approval.

Use synthetic starter data where relevant. Teams may create additional synthetic supplier spreadsheets, documents, images, taxonomy, channel rules, regulatory attributes and correction feeds plus representative edge cases, corrections, failure responses and approval states needed to demonstrate the complete solution. No live confidential enterprise data is required.

End-to-End Journey

1. Supplier data received
2. Content parsed
3. Attributes extracted
4. Conflicts resolved
5. Taxonomy mapped
6. Rules validated
7. Gaps enriched
8. Reviewer approved
9. Product published

National Finale Inject

A supplier sends a corrected specification after content has been prepared for several channels. The system must propagate the correction and identify every affected field and channel.

06. Omnichannel Customer Journey Investigation & Recovery Agent

Business Context & Why It Matters

A customer issue can cross website, app, store, fulfilment, payment and contact-centre systems, leaving each team with only part of the story.

The operating environment spans web and app events, store transactions, orders, fulfilment updates, payments, contact-centre records and returns. Evidence can arrive asynchronously, be corrected later or conflict across systems, so the solution must preserve state, provenance and prior actions as the case evolves.

The Challenge & Scope

Reconstruct the full customer journey, identify where it failed and coordinate an appropriate recovery without duplicate promises.

The outcome should be an evidence-backed operational workspace that can progress the journey, surface uncertainty and coordinate authorized actions while preserving accountability. High-value compensation and exceptional remedies follow authorized approval rules.

Core Solution Capabilities

- Assemble and reconcile web and app events, store transactions, orders, fulfilment updates, payments, contact-centre records and returns into a coherent, time-aware operating view.
- Detect missing, stale, duplicated or contradictory information and make uncertainty explicit.
- Dynamically select the next useful evidence, analysis, coordination or permitted action based on the current state and constraints.
- Maintain source provenance, human intervention, approvals and a complete audit trail through the full journey.

End-to-End Capability Expectations & Controls

- Build a durable domain model linking all journey evidence while preserving late, corrected and conflicting versions.
- Determine the next-best evidence request or permitted action dynamically.
- Track long-running workflow state, deadlines and dependencies while preventing duplicate requests, transactions or external actions.

- Represent competing interpretations and show how evidence changes each option.
- Maintain versioned state and separate recorded fact, AI inference, user input, automated action and human decision.
- Coordinate approved components with permission checks, retries, timeouts, idempotency and safe recovery.
- Support intervention, approval gates, audit replay and configurable action boundaries.

Production-Ready Solution & Data Environment

The solution should ingest representative web and app events, store transactions, orders, fulfilment updates, payments, contact-centre records and returns, execute the complete journey, react safely to late-arriving or changed information, recover from partial failure and retain a complete inspectable audit trail.

Use synthetic starter data where relevant. Teams may create additional synthetic journey records and representative edge cases, corrections, failure responses and approval states. No live confidential enterprise data is required.

End-to-End Journey

1. Issue reported
2. Journey assembled
3. Failure located
4. Evidence checked
5. Impact assessed
6. Recovery options ranked
7. Actions coordinated
8. Customer updated
9. Outcome retained

National Finale Inject

A fulfilment status changes after compensation has been proposed but before it is issued. The solution must reassess the recovery plan and prevent conflicting actions.

07. Autonomous AI Workforce Observability & Governance Command Centre

Business Context & Why It Matters

A real-time supervisory control plane that observes every agent, understands behaviour and outcomes, enforces boundaries, intervenes proportionately and continuously improves the autonomous AMS workforce.

As autonomous and semi-autonomous agents perform more AMS work, an agent can repeatedly fail, act outside approved scope, use unauthorised tools or data, collaborate incorrectly, propagate bad context or continue acting while confidence and performance deteriorate.

- **Trust and safety:** Continuous supervision increases confidence that agents are not operating as black boxes.
- **Governance at scale:** Consistent policy across large fleets avoids equivalent growth in manual supervision.
- **Productivity and economics:** Performance, rework, token, tool and runtime cost reveal value per completed outcome.

Complexity layer: Telemetry may be missing, delayed or privacy-

restricted; agent versions and policies change; performance varies by task type; interactions can create cascading risk; and an apparently successful task can fail downstream validation. The system must preserve versions and reason under uncertainty.

The Challenge & Scope

Build a real-time command centre that combines agent identity, task lifecycle, reasoning and decision events, tool calls, scope, policy, performance, cost, inter-agent interactions and business outcomes into one evidence-driven supervisory layer.

Provide fleet and Agent 360 visibility, detect underperformance, unknown anomalies, scope and policy violations, risky interactions and unsupported outcomes; coach agents that can improve; proportionately constrain or stop agents that create unacceptable risk; and measure whether intervention improved the workforce.

- **Observe:** Capture identity, tasks, plans, evidence, tools, actions, handoffs, policies, cost and outcomes.
- **Understand:** Build behavioural baselines by agent, version, task and operating context.
- **Detect:** Identify drift, scope crossing, confidence collapse, quality failure, loops and cost anomalies.
- **Govern:** Evaluate every material action against scope, autonomy tier, policy and contextual constraints.
- **Intervene:** Observe, coach, constrain, human-gate, pause, quarantine, terminate, roll back or escalate.
- **Validate and Improve:** Challenge self-reported success, verify business outcomes and show before/after performance.

Coordinated Supervisory Agents

Agent	Expected intelligence	Output
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Behaviour Intelligence	Baselines action sequences, tool use, latency and outcomes	Anomaly and drift evidence
Performance Evaluator	Multidimensional effectiveness by task and version	Performance assessment
Policy & Scope Guardian	Evaluates action against policy and autonomy	Allow, gate or block
Risk Scoring	Combines anomaly, reversibility and blast radius	Dynamic agent/task risk
Interaction Graph	Detects loops, risky chains and bad-context propagation	Interaction warnings
Supervisor & Intervention	Selects and executes proportionate governed response	Coach, constrain, pause or escalate
Validation & Sufficiency	Checks actual outcome and whether context could support it	Outcome and context verdict
Explainability & Command	Explains evidence and supports governed natural-language commands	Audit-ready explanation

Finale Conditions

Multiple agent risks collide after work has already propagated:

1. **Underperformance drift:** A Log Analytics Agent drops from its historical success baseline while retries, latency and cost rise.
2. **Scope crossing:** An Access Governance Agent attempts a write outside the approved service boundary.
3. **Circular delegation:** Agents A, B and C reinforce an incorrect conclusion that has already reached a remediation agent.
4. **Confidence collapse:** The remediation agent continues proposing high-impact actions below the approved confidence

threshold.

5. **Insufficient context:** A Knowledge Agent confidently answers from evidence that cannot support the conclusion.
6. **Business pressure:** A critical-service backlog grows while quarantine would reduce fleet capacity.

Expected response:

- Distinguish drift, policy breach, risky interaction, insufficient context and outcome failure.
- Coach the degraded agent, block scope crossing, break the loop and human-gate high-impact actions.
- Preserve healthy work, pause only impacted tasks and reroute critical backlog to trusted peers.
- Show risk factors, applicable policy, confidence, context sufficiency and blast radius.
- Canary revised agents before restoring autonomy and reconcile partial actions.
- Demonstrate before/after performance and validated outcome following coaching or control.

Human Decision and Action Boundaries

Supervisory AI may observe, score risk, detect anomalies, issue low-severity coaching, recommend controls, insert preauthorised validation, automatically block explicitly prohibited actions, reduce preapproved permissions, pause a task under defined kill-switch policy, and draft evidence-backed explanations.

Authorised people must own governance policy changes, exceptions, restoration of quarantined high-criticality agents, acceptance of residual risk, production-agent termination, irreversible rollback, sensitive access-control modifications, and external communications unless explicitly delegated.

The system governs agents and tasks, not people.

End-to-End Capability Expectations

- Build a durable agent-workforce graph linking identities, versions, tasks, reasoning events, tools, policies, delegations, interventions, outcomes and business services.
- Evaluate performance contextually by task type, criticality and version rather than one global leaderboard score.
- Detect seeded policy violations and unknown behavioural anomalies without tailoring logic only to test cases.
- Separate insufficient context, reasoning error, tool failure, policy breach and downstream validation failure.
- Reconcile agent self-reported success with independent technical and business evidence.
- Detect circular delegation, collusion-like reinforcement and the original source of cascading bad context.
- Apply the least-disruptive effective intervention, preserve valid completed work and support rollback, rerouting and human takeover.
- Attribute drift and violation to agent, model, prompt, tool and policy versions.
- Handle missing or privacy-restricted traces gracefully and show uncertainty.
- Provide tamper-evident audit replay of action, detection, decision, approval, enforcement and final outcome.

Proportionate Intervention Model

Level	Control	Expected use
0	Observe	Increase monitoring; no intervention
1	Coach	Recommendation or warning with workflow acknowledgement
2	Constrain	Reduce tools/actions, autonomy or require validation

3	Human Gate	Require approval before consequential action
4	Pause / Quarantine	Stop new tasks and isolate pending review
5	Terminate / Roll Back	Stop task/session and compensate where supported
6	Escalate	Notify governance/owner and recommend version remediation

Minimum Finale Demonstration

Fleet view plus Agent 360; one underperformance coaching flow; one unknown anomaly; one scope violation blocked; one circular agent chain interrupted; one false self-reported success challenged; and one measurable improvement after intervention.

Production-Ready Solution & Data Environment

Representative data:

- 128 agents with role, owner, model, version, scope, tools, criticality and autonomy tier.
- 2,000–5,000 tasks with lifecycle, retries, plans, confidence, evidence and validation.
- Tool calls, action class, side effects, authorisation, scope rules, thresholds and kill-switch policy.
- 30–90 days of success, quality, latency, override, rollback, cost and rework signals.
- Delegation graph, circular chains, bad-context propagation, violations and anomalies.
- Service criticality, journeys, SLOs, blast radius, human ratings and corrections.

Use synthetic data only. Protect prompts, context, reasoning artefacts, secrets and sensitive parameters through minimum-necessary capture, authorised access, redaction and retention controls.

End-to-End Journey

1. Fleet registered
2. Task observed
3. Behaviour compared
4. Policy evaluated
5. Interactions traced
6. Outcome challenged
7. Risk synthesised
8. Control selected
9. Human gate applied
10. Work reconciled
11. Improvement measured
12. Audit sealed

National Finale Inject

The Supervisory Cascade: underperformance drift, scope crossing, circular delegation, confidence collapse, insufficient context and business pressure occur together after work has already propagated. The system must separate the failure modes, apply proportionate controls, preserve healthy work, recover safely and prove measurable improvement.

08. Autonomous Major Incident Intelligence Command Centre

Business Context & Why It Matters

Continuously understand live human and machine signals, maintain an evidence-driven incident picture, guide safe recovery and independently validate technical, service and business restoration.

Major incidents are managed through fragmented conference bridges, team chat, monitoring dashboards, ITSM records, runbooks, change records, topology, knowledge articles and individual SME experience. The result is high cognitive load, parallel or repeated troubleshooting, weak decision memory and avoidable recovery delay.

- **Information fragmentation:** Operational truth is distributed across human conversation and machine evidence.
- **Unsafe recovery pressure:** Teams can repeat failed actions, act on weak assumptions or declare recovery too early.
- **Business impact is obscured:** Technical symptoms do not always show affected journeys, customers, regions, revenue or residual risk.

Complexity layer: Evidence may be late, duplicated, stale, corrected or contradictory. A remediation may be partially executed, an action may have completed despite a timeout, and technical health may recover before customers or business services. The system must preserve all versions and perform targeted re-reasoning when material evidence changes.

The Challenge & Scope

Build an AI-powered command centre that listens to the major-incident bridge, reads collaboration chat and operational signals, continuously updates its understanding, ranks competing root-cause hypotheses, recommends the next best diagnostic or recovery action, prevents repetition of past mistakes, supports the Incident Commander and accelerates safe service restoration.

During a simulated P0/P1 production incident, the solution must ingest

synchronised human and machine signals, construct a trusted situation and impact model, continuously challenge its own understanding, coordinate authorised diagnostics and recovery, communicate through role-specific views and validate recovery across technical, service, customer and business layers.

- **Observe:** Continuously consume bridge, chat, alerts, metrics, logs, traces, changes, topology and business signals.
- **Understand:** Separate facts, user statements, assumptions, hypotheses, decisions, commitments, blockers and contradictions.
- **Reason:** Maintain at least three plausible causes, evidence for and against, confidence movement and unresolved questions.
- **Recommend:** Select the next diagnostic or recovery step using impact, information gain, urgency, risk, permissions and dependencies.
- **Coordinate:** Orchestrate authorised agents, tools, approvals, communications, retries, hand-offs and human takeover.
- **Validate:** Confirm technical, service, customer and business restoration through an appropriate stability window.

Coordinated Agent Model

Agent	Required intelligence	Primary output
Conversation Understanding	Interprets bridge/chat and contradictions	Structured incident facts and events
Telemetry Intelligence	Correlates metrics, logs, traces and alerts	Anomalies and evidence
Topology / Dependency	Understands resource, service and customer dependencies	Blast radius and affected path
Change Intelligence	Links onset with release, configuration and change activity	Change-risk evidence

Historical Learning	Finds similar incidents, prior fixes and mistakes	Lessons and warnings
Hypothesis Reasoning	Creates competing hypotheses and updates confidence	Ranked causes
Next-Best-Action	Plans diagnostic or recovery step and alternatives	Action plan, risk and fallback
Safety / Governance	Checks policy, permissions, confidence and risk	Allow, approve or block
Communication	Produces role-specific messages grounded in state	Approved-ready drafts
Validation	Tests technical and business restoration	Layered recovery verdict
Incident Governor	Coordinates agents, resolves conflicts and maintains memory	Coherent evolving plan

Finale Conditions

A configuration rollback is approved and partially executed. Then:

1. One region reports success while another tool times out, leaving external execution state uncertain.
2. New topology evidence reveals an aggregation failure affecting multiple regions.
3. Failover alters service dependencies and invalidates part of the earlier blast-radius model.
4. A previously comparable incident used topology that no longer matches the current route.
5. An executive update already attributes the incident primarily to configuration.

The solution must preserve valid completed work, pause only invalidated actions, explain changed assumptions, determine what completed before retrying, re-rank configuration and fibre hypotheses, recompute blast radius, lower the relevance of obsolete history, produce revised communications with approval, and validate layered recovery.

Human Decision and Action Boundaries

AI may perform or prepare read-only diagnostics, correlation, evidence requests, timeline construction, hypothesis ranking, next-best-action recommendations, role-specific communication drafts, and explicitly permitted low-risk checks with complete logging.

Authorised people must own incident strategy, priority, declaration, severity changes, risk acceptance, closure, external and executive communications, restart, rollback, traffic shift, isolation, configuration change, emergency override and customer-impacting actions.

End-to-End Capability Expectations

- Build a durable incident graph linking alarms, symptoms, services, resources, customers, changes, actions, approvals, evidence and outcomes without losing superseded versions.
- Reconcile timestamps, duplicate alert storms, missing telemetry and conflicting statements using source authority, freshness, confidence and field-level provenance.
- Maintain competing service-impact and root-cause hypotheses and seek discriminating evidence.
- Select next-best evidence or permitted action using urgency, business impact, information gain, risk, permissions and dependencies.
- Propagate downstream impact through topology, customer journeys and services.
- Coordinate approved diagnostics with preconditions, checkpoints, timeouts, retries, idempotency, approval expiry and human

takeover.

- Reconcile uncertain external state before retrying an action that might already have completed.
- Support partial failure, compensation, rollback and resumability while preserving valid work.
- Separate technical, service, customer, business and stability-window recovery.
- Retain verified learning without silently converting an unverified hypothesis into future guidance.

Minimum Finale Demonstration

A live evolving incident with bridge and chat, at least three hypotheses, visible confidence shift, historical mistake prevention, one human-approved multi-step recovery, partial-execution reconciliation, role-specific communication and layered recovery validation.

End-to-End Journey

1. Signals received
2. Incident clustered
3. Situation built
4. Blast radius modelled
5. Hypotheses ranked
6. Next evidence chosen
7. Action governed
8. Recovery coordinated
9. State reconciled
10. Communication approved
11. Recovery validated
12. Audit and learning sealed

National Finale Inject

The Recovery Collision: a configuration rollback is approved and partially executed when new topology evidence, a traffic-path change and superseded historical context invalidate part of the leading view. The system must reconcile uncertain execution state, re-rank hypotheses and blast radius, revise communications with approval and independently validate layered recovery.

09. AI-Powered Application Security Detection, Remediation & Autonomous Validation

Business Context & Why It Matters

Modern enterprise applications undergo continuous development and frequent releases, making it increasingly difficult for security teams to manually identify, validate, prioritize, remediate and retest vulnerabilities at scale.

Traditional approaches often use separate tools for static analysis, vulnerability assessment, penetration testing, remediation and validation. This creates fragmented workflows, manual effort, false positives, delayed remediation and limited visibility into whether vulnerabilities were resolved.

The Challenge & Scope

Design and build a production-ready AI-powered Application Security Solution covering an end-to-end lifecycle across source-code analysis, controlled penetration validation, remediation and security closure.

1. Analyze application source code and identify potential security vulnerabilities.
2. Classify detected vulnerabilities by Critical, High, Medium or Low severity.
3. Explain location, type, potential impact, severity rationale, evidence and remediation.
4. Understand source code, functional behaviour, design information and test cases to generate penetration scenarios.
5. Execute generated tests only against a controlled local target environment.
6. Correlate source findings and controlled penetration results to assess exploitability.
7. Generate an overall risk score using severity, exploitability, business impact, confidence and correlated findings.
8. Recommend source-code fixes.
9. Provide human review to approve, reject or modify proposed fixes.
10. Apply approved code changes automatically with annotations or comments.
11. Re-analyze modified code and re-execute applicable tests.
12. Verify remediation and whether Critical or High vulnerabilities remain.
13. Maintain an end-to-end evidence and audit trail.

Expected AI Capabilities

- Understand source-code context and application behaviour.
- Identify security-relevant code patterns.
- Reason about severity and exploitability.
- Distinguish meaningful findings from lower-risk or false-positive findings.
- Generate security test scenarios dynamically.
- Correlate static and runtime findings.
- Prioritize vulnerabilities using contextual risk.
- Generate contextual remediation recommendations and secure code modifications.
- Explain recommendations and decisions.
- Incorporate human approval/rejection feedback.

- Determine what should be retested after remediation.

Technical Expectations

- Use one or more available LLMs with suitable security-analysis technologies.
- Demonstrate multi-language source-code security analysis.
- Perform vulnerability classification and prioritization.
- Generate and execute controlled tests against the sandbox target.
- Assess exploitability and confidence.
- Generate unified application risk scoring.
- Support human approval before code change.
- Apply approved changes, rescan, retest and provide evidence-based closure.
- Support at least two programming languages or application technology stacks.

Production-Ready Solution & Data Environment

Participants should create or use synthetic applications, deliberately vulnerable applications, sample source-code repositories, synthetic test cases and synthetic application data.

Penetration testing must operate only against a locally deployed or otherwise controlled sandbox application created for the challenge. No live customer, TCS, internet-facing, production, corporate or unauthorized external system may be targeted.

End-to-End Journey

1. Application Intake
2. AI Security Assessment

3. Security Test Generation
 4. Controlled Validation
 5. Unified Risk Assessment
 6. AI Remediation
 7. Human Validation
 8. Automated Correction
 9. Security Retesting
 10. Final Security Status
 11. Evidence and Audit
 12. Outcome Metrics
-

10. AI-Powered Insurance Decision & Regulatory Intelligence Platform

Business Context & Why It Matters

Insurance operations frequently require experts to interpret large volumes of unstructured documents, policies, business rules, regulatory requirements and historical precedent before making a defensible decision.

The challenge is not simply to summarize information, but to identify material evidence, detect gaps and inconsistencies, apply applicable rules, explain the basis of a decision, and produce documents needed to move the case forward.

This challenge combines two reusable intelligence journeys: **Subrogation Recovery Decisioning** and **Rate Filing & Regulatory Readiness**.

The Challenge & Scope

Develop a production-ready AI solution that interprets complex insurance evidence, policies, regulations and historical precedents to support defensible business decisions. It must identify relevant evidence and rules, detect missing or conflicting information, recommend an appropriate action, generate supporting documentation, provide traceable explanations, and maintain human oversight and auditability.

Scenario A: Subrogation Recovery Decisioning

Business question: Is there a valid recovery opportunity, against whom, why, what evidence supports the conclusion, what evidence is missing or conflicting, and what should happen next?

Expected capabilities:

- Reconstruct the incident and identify key entities, events and relationships.
- Interpret relevant policy and coverage language.
- Reason about liability using supplied evidence and business rules.
- Identify a potentially responsible third party, carrier, manufacturer, contractor or owner where supported.
- Assess evidence sufficiency and identify missing, weak or conflicting evidence.
- Evaluate recovery paths using explicit rules, costs, timelines and assumptions.
- Calculate recovery economics using deterministic calculations.
- Recommend Recovery Recommended, Additional Evidence Required, or Do Not Pursue.
- Generate a recovery brief or demand package for human review.

Scenario B: Insurance Rate Filing & Regulatory Readiness

Business question: Is the filing ready for regulatory submission, what is missing or inconsistent, which requirements apply, what questions could

reasonably arise based on rules or precedent, and how should the submission be explained?

Expected capabilities:

- Interpret actuarial outputs and the proposed rate change.
- Determine applicable state/product filing requirements.
- Retrieve and cite relevant requirements, guidance and precedent.
- Check completeness and identify missing mandatory information.
- Perform cross-document consistency and validation checks.
- Identify potential regulator questions or objections.
- Generate explanations and supporting content for human review.
- Produce a regulator-ready or SERFF-ready package without live SERFF integration.
- Recommend Ready, Conditionally Ready, or Not Ready.

Reusable Intelligence Platform

1. **Ingest:** Documents, structured data, policies, rules and evidence.
2. **Understand:** Extract entities, facts, relationships and material clauses.
3. **Retrieve:** Find relevant policy, regulatory requirement, rule or precedent.
4. **Reason:** Resolve evidence, identify contradictions, apply rules and assess gaps.
5. **Act:** Recommend next action, generate documents, obtain approval and retain audit trail.

Mandatory Functional Expectations

- Multi-document understanding
- Evidence-grounded reasoning
- Contradiction and gap detection
- Deterministic business-rule execution
- Document generation
- Human-in-the-loop controls

- Traceability and audit
- Explainability

Data Considerations

Use synthetic data. A small universe may cover two or three states, selected products, limited regulatory documents, historical objections, and eight to ten claim/evidence documents. Include missing information, contradictory evidence and edge cases.

National Finale Inject

New evidence or rules are introduced during evaluation. A new witness statement may contradict the police report, a policy endorsement may arrive, a regulatory bulletin may change a requirement, or the proposed rate change may be modified. The solution must identify the impact and re-evaluate affected decisions, validations, explanations and generated documents.

11. Autonomous Customer Retention Intelligence & Save Assurance Command Centre

Business Context & Why It Matters

Retention programmes are often reactive and generic. Agents lack a consolidated view of churn risk, customer value, likely reasons and the most suitable intervention. The operating picture may span usage and

recharge history, bills and payments, product holdings, complaints, network experience, campaign history, customer profile, offer catalogue and contact outcomes.

A churn score alone does not explain what changed, which evidence is current, whether a customer is contactable, which offer is eligible, or whether an intervention improved the relationship. Signals may be delayed, duplicated, stale, corrected or contradictory.

The Challenge & Scope

Build an AI-powered Customer Retention Intelligence & Save Assurance Command Centre that continuously fuses customer, commercial, service and interaction signals into a trusted retention picture; estimates customer and revenue risk; maintains competing churn-cause hypotheses; recommends safe next-best evidence and interventions; coordinates approved outreach and fulfilment workflows; and validates customer save, experience and value outcomes.

Core Solution Capabilities

- Consume usage, recharge, billing, payment, complaint, network, campaign and interaction events.
- Maintain customer value, products, risk, facts, assumptions, hypotheses, evidence, interventions, consent, owners, risks and unresolved questions.
- Connect network experience, product journeys, household/ account relationships and revenue exposure.
- Generate at least three plausible churn causes, show confidence and evidence for or against, and explain movement or rejection.
- Recommend the next check, conversation, service recovery, offer or non-offer action with rationale, eligibility, risk, prerequisites and fallback.
- Retrieve comparable journeys and warn when an intervention previously failed or accelerated churn.

- Coordinate approved contact, offer reservation and fulfilment with consent and approval controls.
- Declare a save only after behavioural, service, commercial and customer-response signals meet defined criteria through an observation window.

Human Decision and Action Boundaries

Retention leaders own strategy, policy, budgets, fairness and outcome definitions. Retention agents own customer conversation, judgement, disclosures and final offer acceptance. Commercial concessions, contract changes, credits, waivers, plan migrations and customer-impacting fulfilment require authorized approval. Consent, privacy-sensitive use, vulnerable-customer handling and regulated communication require explicit controls and human review.

End-to-End Capability Expectations

- Build a durable customer-retention graph linking account, products, usage, payments, complaints, network experience, campaigns, offers, interactions, consent, actions, evidence and outcomes.
- Reconcile duplicate events, missing usage, stale profiles, corrected payments and conflicting statements with provenance.
- Separate observed fact, customer statement, AI inference, proposed intervention, automated step and human decision.
- Maintain competing causes and seek discriminating evidence.
- Estimate downstream impact across products, household relationships, service journeys, value and contact fatigue.
- Select next-best evidence or action using urgency, value, information gain, benefit, fairness, policy, cost, permissions and dependencies.
- Coordinate approved agents with reservations, checkpoints, retries, timeouts and idempotency.
- Prevent duplicate incentives and conflicting messages.

- Support partial failure, compensation, withdrawal, takeover and resumability.
- Independently validate save, experience and value outcomes.

End-to-End Journey

1. Signals received
2. Risk detected
3. Impact modelled
4. Causes ranked
5. Evidence requested
6. Action approved
7. Intervention coordinated
8. Save validated

National Finale Inject

A retention offer and plan migration have been approved, the offer has been reserved, and outreach has been partially completed. New network and complaint evidence then shows that the leading price-sensitivity hypothesis is insufficient, while a corrected payment event invalidates part of the affordability evidence. The command centre must preserve valid work, pause only invalidated steps, reconcile state, prevent duplicate incentives or conflicting messages, re-rank causes and produce a targeted revised plan.

12. Autonomous Media Metadata Trust & Editorial Governance Command Centre

Business Context & Why It Matters

TV and digital distribution depend on reliable programme identity, synopsis, cast, crew, genre, ratings, rights and channel-specific descriptions. Manual research is slow, inconsistent and difficult to scale. Weak metadata can impair search, recommendation, scheduling, compliance checks and downstream publishing.

Programme titles, seasons, episodes and people may be ambiguous. Sources may disagree, repeat outdated credits, omit fields or correct facts after approval. Channel policies differ by audience, market and text length. The system must maintain identity and provenance, distinguish fact from inference, manage conflicting candidates, preserve superseded versions and trigger targeted re-evaluation when material evidence changes.

The Challenge & Scope

Build an AI-powered Media Metadata Trust & Editorial Governance Command Centre that continuously assembles approved corpus evidence into a versioned programme record, resolves entity and fact conflicts, maintains competing metadata candidates, recommends the next evidence or editorial action, coordinates approval and publication workflows, and validates channel outputs after distribution.

Core Solution Capabilities

- Ingest schedules, title IDs, existing metadata, approved articles/catalogues, trailers and images; extract claims with source and timestamp.
- Link titles, variants, seasons, episodes, people, roles, franchises, territories and source records while retaining ambiguity.
- Maintain multiple field candidates, confidence and evidence for/against.

- Recommend the next source, targeted question, escalation or approval step using information gain, criticality and deadline.
- Generate descriptions of different lengths and styles without changing approved facts.
- Enforce taxonomy and prohibited-content rules.
- Coordinate review, rights/compliance checks, publication, correction, rollback and re-approval.
- Validate completeness, consistency, identity, length, taxonomy and post-publication parity.
- Record evidence, AI inferences, editorial decisions, publications, corrections and verified lessons.

Human Decision and Action Boundaries

Editors remain accountable for factual acceptance, tone, taxonomy exceptions and publication approval. Rights, legal, compliance, age-rating and market-sensitive classifications require authorized human decision. AI must not silently overwrite approved metadata, merge identities, publish corrections or retract live records.

End-to-End Capability Expectations

- Build a durable metadata evidence graph linking identity, field claims, sources, assets, policies, approvals, channel variants, publication events and outcomes.
- Reconcile duplicate feeds, aliases, transliterations, reused titles, season/episode ambiguity, stale catalogues and contradictory claims.
- Separate source claim, extracted fact, AI inference, proposed value, approved value, published value and correction.
- Maintain at least three plausible candidates where ambiguity exists.
- Propagate changes across dependent fields and channel variants while isolating unaffected content.
- Coordinate research, entity resolution, policy, editorial, publication

and validation with permissions, retries and idempotent publication keys.

- Reconcile uncertain external publication state before retrying.
- Support rejection, partial approval, compensation, rollback, takeover and resumability.
- Block generated copy that introduces unsupported facts, identity leakage or policy violations.
- Independently validate master data and destination output.

End-to-End Journey

1. Feed received
2. Identity candidates built
3. Evidence reconciled
4. Metadata proposed
5. Editor reviews
6. Publication approved
7. Channels updated
8. Outcome validated

National Finale Inject

A master record for a newly acquired drama has been approved and publication has partially completed to two channels. A late rights-holder correction shows that the apparent lead actor belongs to a different programme with the same title, while a revised trailer contradicts the approved genre and synopsis. The command centre must preserve valid fields, pause invalidated publication actions, reconcile destination state, re-rank candidates, obtain required approvals and retain a visible explanation of what changed.

13. Autonomous Supplier

Commitment Assurance & Recovery Control Tower

Business Context & Why It Matters

Planners manage weekly supplier commitments across many parts, suppliers and manufacturing sites using multiple systems and manual reports. Late visibility into commitment changes can create shortages, unstable production plans, expedites and delayed decisions.

Commitments are revised by week, split across sites and constrained by lead time, logistics, capacity, quality and allocation. Feeds may be duplicated, stale, corrected or contradictory. A supplier may increase its commit while logistics evidence shows the material cannot arrive.

The Challenge & Scope

Build an AI-powered Supplier Commitment Assurance & Recovery Control Tower that continuously fuses supplier commits, demand, inventory, production plans, logistics, quality and planner signals into a versioned supply situation model; maintains competing risk and cause hypotheses; recommends safe next-best mitigation; coordinates approved recovery workflows; and validates coverage, delivery and production outcomes.

The solution must not be a passive chatbot, static search experience or dashboard-only prototype. It must sense, reconcile, reason, challenge, recommend, coordinate, validate and remember as evidence and workflow state change.

Core Solution Capabilities

- Ingest weekly commits, forecasts, orders, inventory and plan changes; normalize versions, units, dates and source authority.
- Maintain SKU-site-week coverage, shortages, excess, assumptions, versions, hypotheses, actions, owners, risks and unresolved questions.
- Connect parts, suppliers, lanes, sites, products and customer demand; show confirmed and predicted downstream impact.
- Rank causes such as capacity, lead-time, logistics, quality, allocation or data error with evidence for and against.
- Recommend evidence checks and actions with quantity, timing, impact, cost/risk, prerequisites, approval and fallback.
- Compare expedite, alternate source, inventory transfer, substitution, resequencing and controlled demand allocation.
- Coordinate supplier, logistics, quality and plant workflows with checkpoints, retries, timeouts and takeover.
- Verify actual receipts, usable material, restored coverage and production recovery.

Human Decision and Action Boundaries

Supply planners own demand prioritisation, allocation and production-plan acceptance. Procurement and commercial owners approve supplier commitments, purchase-order changes, premium freight and contractual actions. Site and quality leaders approve substitution, deviation, inspection release and sequencing changes. External communication, financial commitments and customer-impacting decisions require human review and approval.

End-to-End Capability Expectations

- Build a durable supply assurance graph linking supplier, part, site, week, demand, inventory, purchase order, shipment, quality state, production dependency, actions, evidence and outcomes.
- Reconcile unit conversions, calendar boundaries, duplicate schedules, stale EDI extracts, corrected commits, split shipments

and conflicting statements.

- Separate received commitment, planner statement, model forecast, inferred shortage, proposed action, approved action, executed action and verified outcome.
- Maintain at least three plausible variance or delivery-risk hypotheses where uncertainty exists.
- Propagate downstream impact to assemblies, plants, production orders, customer demand and service commitments.
- Select next-best evidence or mitigation using urgency, shortage magnitude, criticality, information gain, lead time, cost, risk, permissions and dependencies.
- Coordinate approved workflows with preconditions, quantity reservations, permissions, checkpoints, retries, timeouts and idempotency.
- Reconcile uncertain external state before retrying a purchase-order change, expedite or allocation.
- Support partial execution, compensation, rollback, takeover and resumability.
- Validate recovery across shipment, receipt, quality release, usable inventory, coverage and production stability.

End-to-End Journey

1. Commits received
2. Variance detected
3. Impact modelled
4. Causes ranked
5. Scenarios compared
6. Action approved
7. Recovery coordinated
8. Coverage validated

National Finale Inject

An alternate-source expedite and production resequencing plan has been approved and partially executed for a critical SKU. The strategic

supplier then increases its weekly commit, but a new port-closure event and a quality-hold update show that much of the apparent recovery cannot reach or be released at the required plant. The control tower must preserve valid reservations and completed work, pause only invalidated actions, reconcile the changed state, prevent double ordering, re-rank risk hypotheses and propose a targeted revised mitigation plan.

14. Wealth Management Portfolio Intelligence & Advisory Workspace

Business Context & Why It Matters

Wealth managers must continuously connect a client's goals, risk profile, investment mandate, holdings, cash flows, product information, research, market events, tax or liquidity constraints and prior conversations. These inputs change at different speeds and are usually spread across multiple systems and documents.

The difficult part is not simply summarizing a portfolio. It is maintaining a trustworthy, current view of the client situation, identifying when suitability or portfolio assumptions have changed and preparing evidence-backed advisory options without allowing AI to make an unapproved investment decision or execute a trade independently.

The Challenge & Scope

Build an AI-powered wealth-management workspace that assembles the

complete client and portfolio context, detects material changes, identifies concentration or suitability concerns, prepares explainable advisory alternatives and coordinates the next appropriate review or action.

The solution should help a relationship manager or investment specialist understand why an option is being proposed, what evidence supports it, which constraints apply, what remains uncertain and what requires client consent or authorized human approval. Investment advice, suitability sign-off and trade execution remain with authorized people and systems.

Core Solution Capabilities

- Assemble and reconcile client goals, risk profile, investment mandate, holdings, transactions, cash flows, product data, research, market information, constraints and interaction history into a coherent time-aware view.
- Detect missing, stale, duplicated or contradictory information and make uncertainty, data lineage and suitability gaps explicit.
- Identify material portfolio changes, concentration or drift and prepare explainable advisory alternatives linked to the client objective and constraints.
- Maintain source provenance, client consent, human approvals and a complete evidence and action trail.

End-to-End Capability Expectations

- Build a durable client-and-portfolio model linking goals, risk profile, mandate, holdings, transactions, products, research, market events, constraints, communications, recommendations and approvals.
- Test whether evidence is current and sufficient and trigger targeted reevaluation when a material fact changes.
- Generate and compare multiple advisory alternatives with impact, constraints, evidence, trade-offs and unresolved assumptions.
- Apply deterministic suitability, eligibility, concentration and

mandate checks alongside AI reasoning.

- Separate recorded facts, calculated measures, AI inference, advisor input, client instruction, human approval and executed transaction.
- Coordinate approved services with permissions, retries, timeouts, idempotency and safe recovery.
- Support role-based access, client-consent checkpoints, approval gates and audit replay.

Production-Ready Solution & Data Environment

Use synthetic client profiles, goals, risk assessments, mandates, holdings, transactions, cash flows, product attributes, research notes, market events, tax or liquidity constraints, communications, recommendation histories, approvals and representative failure responses. No live client, investment or confidential banking data is required.

End-to-End Journey

1. Client context assembled
2. Portfolio reconciled
3. Goals and constraints validated
4. Drift and risks detected
5. Alternatives prepared
6. Suitability evidence checked
7. Advisor reviews
8. Client decision captured
9. Outcome and evidence retained

National Finale Inject

After an advisory plan has been prepared and partially approved, a

material change in the client's liquidity need and a sharp market move alter portfolio risk and invalidate part of the original rationale. The solution must identify exactly what changed, reassess affected options, prevent stale recommendations from progressing and make the revised evidence and decisions visible to the advisor.

15. AI-Driven Cost of Care Optimization & Population Health Intervention

Business Context & Why It Matters

Build a production-ready AI solution that detects early signals of avoidable healthcare cost, understands the clinical and contextual drivers behind those signals, and recommends timely interventions that can improve outcomes while reducing total cost of care.

Healthcare spend often rises because intervention happens after a condition becomes acute. Important indicators can be distributed across clinical, pharmacy, behavioral, social and environmental data rather than appearing as one obvious event.

The solution should identify weak signals early and turn them into an actionable sequence:

Detect → Understand → Prioritize → Intervene → Measure Outcome

The Challenge & Scope

1. Detect Meaningful Signals

Identify changes, gaps, trends and non-occurrence of expected actions from structured and unstructured sources:

- Medication refill/adherence gaps
- Missed appointments or care activities
- Remote monitoring deviations
- Behavioral or engagement decline
- Environmental or utility events

2. Understand the Risk Pattern

Connect the signal with the member's clinical condition, treatment dependency, context and likely downstream healthcare event:

- Condition and treatment relationships
- Social and environmental context
- Risk severity and urgency
- Likely avoidable utilization

3. Prioritize the Intervention

Rank situations and explain why intervention is relevant:

- Risk and expected impact
- Intervention urgency
- Member engagement propensity
- Human review thresholds

4. Recommend and Track Action

Recommend who should intervene, what should be done, through which channel and within what timeframe:

- Nurse/care-manager outreach

- Telehealth or escalation
- Medication delivery or reminder
- Transport or alternate care support

End-to-End AI Intervention Journey

- **Sense:** Ingest clinical, pharmacy, behavioral, social and external signals.
- **Detect:** Identify events, trends, gaps and missing expected events.
- **Reason:** Connect condition, dependency, context and downstream risk.
- **Prioritize:** Rank risk, urgency, likely avoidable cost and intervention value.
- **Intervene:** Recommend person, action, channel and timing.
- **Measure:** Track outcome, engagement, avoided utilization and feedback.

Multi-Modal Signal Sources

- EHR and encounter data
- Claims and pharmacy/refill data
- Laboratory and remote-monitoring data
- Clinical notes and member communications
- Weather, air quality and emergency signals
- Utility outages, mobility or transportation context

Key Innovation: Non-Occurrence Detection

The system must treat the absence of an expected event as a signal. Examples include a refill that did not happen, a recurring measurement that was not submitted, an expected appointment that was missed, or an outreach response that did not occur. The solution should explain how

expectations are established and how exceptions are handled.

Human Oversight

Provide review thresholds, exception handling, confidence indicators and a clear mechanism for correcting or overriding recommendations.

Data Approach

Teams may use synthetic, open or self-generated data. No protected health information is required. Data may include Synthea or synthetic members, synthetic claims, refill/adherence events, remote-monitoring samples, weather/AQI, social determinants, utility and mobility events.

End-to-End Journey

1. Sense
 2. Detect
 3. Reason
 4. Prioritize
 5. Intervene
 6. Measure
-

16. Medical Records Intelligence & Longitudinal Health Record Engine

Business Context & Why It Matters

Build a production-ready AI solution that transforms heterogeneous medical documents into trustworthy, structured, standards-aligned information, preserves traceability to the source, connects facts across time, and makes the resulting member record usable for downstream care, risk and analytics.

The business problem is not simply OCR. It is to **understand** → **extract** → **validate** → **correlate** → **route** information without losing clinically meaningful content or inventing unsupported facts.

The Challenge & Scope

Turn fragmented documents into a reliable longitudinal record that can support clinical operations, care coordination, risk analysis, reporting and downstream data products.

Every important extracted fact should be traceable to its source document and location, with confidence and an explicit path for human review.

1. Understand Document Context

- Scanned or native digital input
- Tables, forms and mixed layouts
- Multiple documents combined in one file
- Abbreviations, shorthand and local conventions

2. Extract Clinical Information Losslessly

- Problems, diagnoses and procedures
- Medications and allergies
- Labs, observations and findings
- Dates, units, ranges and contextual qualifiers

3. Normalize and Validate Meaning

- Clinical concept normalization
- Negation and uncertainty handling
- Abbreviation resolution
- Confidence scoring and completeness checks

4. Correlate and Route

- Entity resolution
- Temporal linking
- Duplicate and contradiction detection
- Dynamic routing to relevant marts

End-to-End Journey

1. **Ingest:** Accept heterogeneous document and record inputs.
2. **Understand:** Classify document, quality, structure and strategy.
3. **Read:** OCR / handwriting / layout-aware extraction.
4. **Extract:** Identify clinical entities, facts and qualifiers.
5. **Normalize:** Map meaning to a consistent clinical representation.
6. **Correlate:** Resolve entities and build a longitudinal member view.
7. **Route:** Send trusted data to the relevant downstream destination.

Medical Understanding

Handle clinical vocabulary, abbreviations and acronyms, eponyms, misspellings, negation, uncertainty, units and reference ranges. Standards may include SNOMED CT, ICD-10/ICD-11, LOINC, RxNorm and FHIR-aligned representation.

Lossless Extraction with Provenance

Each material field should retain its original value, normalized representation where applicable, confidence, and document/page/location provenance. The system should identify potential missing facts or unsupported output.

Longitudinal Correlation

Link the same member, provider, encounter, medication and event across records. Order events over time and surface duplicates, contradictions, gaps or missing follow-up results.

Clinical Trust Requirement

The system must distinguish what the document says from what a model may infer. Unsupported clinical facts must not be invented, and ambiguous or low-confidence information should be visibly routed for review.

Data Approach

Use synthetic, open or self-generated data. No real PHI is required. Include poor scans, tables, mixed content, conflicting values and abbreviated clinical language. Scale should be demonstrated through architecture, repeatability and measurable throughput rather than volume alone.

17. Contract Obligation, SLA & Commercial Leakage Monitor

Business Context & Why It Matters

Contracts contain obligations, service levels, credits and notice requirements that must be mapped to real operational evidence to prevent leakage and missed commitments.

The operating environment spans contracts, amendments, SLA records, invoices, service events, credits, notices, renewal dates and owner actions. Evidence can arrive asynchronously, be corrected later or conflict across systems, so the solution must preserve state, provenance and prior actions as the case evolves.

The Challenge & Scope

Extract obligations with provenance, monitor operational evidence and surface potential breaches, missed credits and leakage.

The outcome should be an evidence-backed operational workspace that can progress the journey, surface uncertainty and coordinate authorized actions while preserving accountability. Legal interpretation, contractual notice and material commercial settlement decisions remain human-owned.

Core Solution Capabilities

- Assemble and reconcile contracts, amendments, SLA records, invoices, service events, credits, notices, renewal dates and owner actions into a coherent, time-aware operating view.
- Detect missing, stale, duplicated or contradictory information and make uncertainty explicit.
- Dynamically select the next useful evidence, analysis, coordination or permitted action based on current state and constraints.

- Maintain source provenance, human intervention, approvals and a complete audit trail.

End-to-End Capability Expectations & Controls

- Build a durable domain model linking contracts, amendments, SLA records, invoices, service events, credits, notices, renewal dates and owner actions while preserving late, corrected or conflicting versions.
- Determine the next-best evidence request or permitted action dynamically.
- Track long-running workflow state, deadlines and dependencies while preventing duplicate requests, transactions or external actions.
- Represent competing interpretations and show which evidence supports, weakens or changes each option.
- Maintain versioned state and separate recorded fact, AI inference, user input, automated action and human decision.
- Coordinate approved components with permission checks, retries, timeouts, idempotency and safe recovery.
- Support role-based intervention, approval gates, audit replay and configurable boundaries.

Production-Ready Solution & Data Environment

The solution should ingest representative contracts, amendments, SLA records, invoices, service events, credits, notices, renewal dates and owner actions, execute the complete journey, react safely to changed information, recover from partial failure and retain a complete inspectable audit trail.

Use synthetic starter data and representative edge cases, corrections,

failure responses and approval states. No live confidential enterprise data is required.

End-to-End Journey

1. Contract ingested
2. Obligations extracted
3. Owners mapped
4. Evidence monitored
5. Exceptions detected
6. Impact quantified
7. Action prepared
8. Commercial review
9. Outcome tracked

National Finale Inject

A contract amendment changes an SLA threshold after potential breaches were flagged. The system must re-evaluate each event using the correct effective version.

18. Sales Pipeline, Revenue Risk & Growth Action Copilot

Business Context & Why It Matters

Pipeline and renewal risk is often identified late because account evidence is scattered and subjective status updates are not connected to outcomes.

The operating environment spans CRM opportunities, account interactions, proposals, product usage, support issues, renewals, forecasts, contacts and sales actions. Evidence can arrive asynchronously, be corrected later or conflict across systems, so the solution must preserve state, provenance and prior actions as the case evolves.

The Challenge & Scope

Identify evidence-backed revenue risk, explain drivers and recommend prioritized growth actions with outcome tracking.

The outcome should be an evidence-backed operational workspace that can progress the journey, surface uncertainty and coordinate authorized actions while preserving accountability. Pricing, commercial commitments and forecast sign-off remain with authorized sales leaders.

Core Solution Capabilities

- Assemble and reconcile CRM opportunities, account interactions, proposals, product usage, support issues, renewals, forecasts, contacts and sales actions into a coherent, time-aware operating view.
- Detect missing, stale, duplicated or contradictory information and make uncertainty explicit.
- Dynamically select the next useful evidence, analysis, coordination or permitted action.
- Maintain source provenance, human intervention, approvals and a complete audit trail.

End-to-End Capability Expectations & Controls

- Build a durable domain model linking CRM evidence and preserving late, corrected and conflicting versions.
- Determine the next-best evidence request or permitted action dynamically.
- Track workflow state and dependencies while preventing duplicate requests and repeated actions.
- Represent competing interpretations, plans or hypotheses.
- Maintain explicit separation of recorded fact, AI inference, user input, automated action and human decision.
- Coordinate approved components safely with permissions, retries, timeouts, idempotency and partial-failure recovery.
- Support role-based intervention, approvals and audit replay.

End-to-End Journey

1. Pipeline ingested
2. Account context built
3. Risks detected
4. Drivers explained
5. Actions generated
6. Owners prioritized
7. Actions executed
8. Forecast updated
9. Revenue outcome tracked

National Finale Inject

Customer usage improves while a key executive sponsor leaves after the risk score was reduced. The system must reassess the opportunity using the new evidence.

19. Travel Operations Situation

Intelligence Agent

Business Context & Why It Matters

Operations teams must interpret schedules, weather, facility restrictions, asset availability and customer impact as one rapidly evolving situation.

The operating environment spans schedules, weather events, airport or station notices, asset or crew availability, customer loads, incident reports and operational actions. Evidence can arrive asynchronously, be corrected later or conflict across systems, so the solution must preserve state, provenance and prior actions as the case evolves.

The Challenge & Scope

Create a coherent operational picture, identify cascading impacts and prioritize the decisions or investigations needing attention.

The outcome should be an evidence-backed operational workspace that can progress the journey, surface uncertainty and coordinate authorized actions while preserving accountability. Safety-critical and operational control decisions remain with authorized operations personnel.

Core Solution Capabilities

- Assemble and reconcile schedules, weather events, airport or station notices, asset or crew availability, customer loads, incident reports and operational actions into a coherent, time-aware operating view.
- Detect missing, stale, duplicated or contradictory information and make uncertainty explicit.

- Dynamically select the next useful evidence, analysis, coordination or permitted action.
- Maintain source provenance, human intervention, approvals and a complete audit trail.

End-to-End Capability Expectations & Controls

- Build a durable domain model linking all operational evidence while preserving late, corrected or conflicting versions.
- Determine the next-best evidence request or permitted action dynamically.
- Track long-running workflow state, deadlines and dependencies while preventing duplicate requests, transactions or repeated actions.
- Represent competing interpretations and show how evidence changes each option.
- Maintain versioned state and explicit separation of fact, inference, input, automated action and human decision.
- Coordinate approved components with permissions, retries, timeouts, idempotency and safe recovery.
- Support role-based intervention, approval gates, audit replay and configurable action boundaries.

End-to-End Journey

1. Signals received
2. Situation assembled
3. Impacts connected
4. Risks prioritized
5. Evidence gaps found
6. Options prepared
7. Controllers reviewed
8. Actions tracked
9. Situation refreshed

National Finale Inject

A weather restriction expands to a second hub while an earlier operational assumption is corrected. The system must recompute downstream impact without retaining stale conclusions.

20. Guest Service Failure Reconstruction & Recovery Agent

Business Context & Why It Matters

A guest complaint may be the final symptom of several reservation, room, housekeeping, maintenance or service handoff failures.

The operating environment spans reservations, room status, housekeeping events, maintenance tickets, dining records, staff notes, guest messages and compensation history. Evidence can arrive asynchronously, be corrected later or conflict across systems, so the solution must preserve state, provenance and prior actions as the case evolves.

The Challenge & Scope

Reconstruct the service journey, identify contributing failures and coordinate an evidence-backed recovery plan.

The outcome should be an evidence-backed operational workspace that

can progress the journey, surface uncertainty and coordinate authorized actions while preserving accountability. Material compensation and liability decisions remain with authorized property staff.

Core Solution Capabilities

- Assemble and reconcile reservations, room status, housekeeping events, maintenance tickets, dining records, staff notes, guest messages and compensation history into a coherent, time-aware operating view.
- Detect missing, stale, duplicated or contradictory information and make uncertainty explicit.
- Dynamically select the next useful evidence, analysis, coordination or permitted action.
- Maintain source provenance, human intervention, approvals and a complete audit trail.

End-to-End Capability Expectations & Controls

- Build a durable domain model linking all guest-service evidence while preserving late, corrected or conflicting versions.
- Determine the next-best evidence request or permitted action dynamically.
- Track long-running workflow state, deadlines and cross-system dependencies while preventing duplicate requests, transactions or repeated external actions.
- Represent competing interpretations, plans or hypotheses and show how evidence changes each option.
- Maintain versioned state and explicit separation of recorded fact, AI inference, user input, automated action and human decision.
- Coordinate approved components with permissions, retries, timeouts, idempotency and safe recovery.
- Support role-based intervention, approval gates, audit replay and configurable action boundaries.

Production-Ready Solution & Data Environment

The solution should ingest representative reservations, room status, housekeeping events, maintenance tickets, dining records, staff notes, guest messages and compensation history, execute the complete journey, react safely to changed information, recover from partial failure and retain a complete inspectable audit trail.

Use synthetic starter data and representative edge cases, corrections, failure responses and approval states. No live confidential enterprise data is required.

End-to-End Journey

1. Failure reported
2. Journey reconstructed
3. Evidence joined
4. Contributors identified
5. Impact assessed
6. Recovery options built
7. Staff approved
8. Actions coordinated
9. Guest outcome tracked

National Finale Inject

An unresolved maintenance ticket is discovered to have caused several downstream failures after a recovery offer is prepared. The solution must revise the causal picture and remedy.